Hospital Mapping in Uganda Using Satellite Imagery and Deep Learning *Study in the uNICIFE Program Focused on Healthcare Development in Uganda, Health Facilities in Sub-Saharan Africa*

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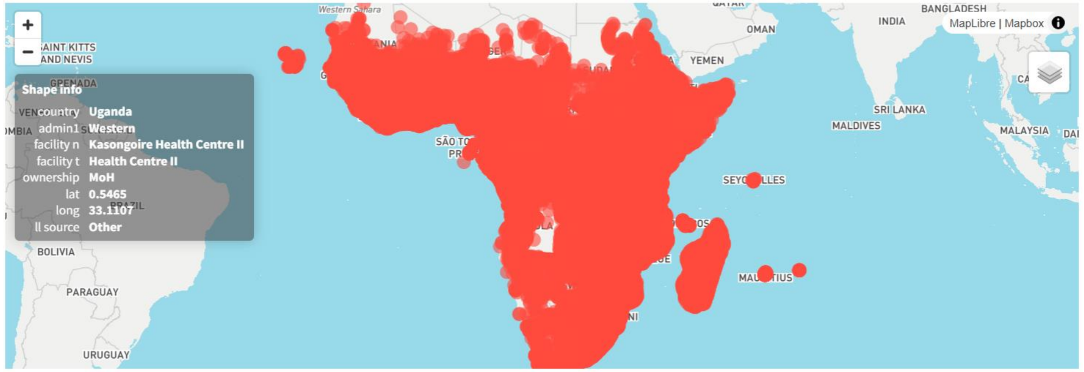
1. Introduction

- Mapping hospitals and healthcare facilities is crucial for improving healthcare accessibility, especially in developing countries like Uganda. With a focus on supporting the health sector, this study uses satellite imagery combined with deep learning techniques to accurately map the location of hospitals in Uganda.
- **Objective:** To develop a reliable method for identifying hospitals in Uganda using publicly available satellite images and deep learning models.
- **Relevance:** This study is part of the uNICIFE program, which focuses on improving healthcare infrastructure in developing countries by providing tools for better resource allocation and health service planning.
- Mapping the spatial distribution of poverty in developing countries presents a significant challenge, both in terms of complexity and cost. The World Bank reports that while poverty has decreased in many regions globally, sub-Saharan Africa continues to struggle with poverty eradication, with several countries still facing substantial challenges (Figure 1).



Background:

Healthcare Challenges in Uganda: Uganda faces significant healthcare access issues, especially in rural areas where hospitals and health facilities are sparse or hard to identify.

Importance of Mapping: By accurately mapping healthcare facilities, Uganda can enhance healthcare delivery, optimize resource distribution, and support future health policy decisions.

Data Collection:

- **Satellite Imagery:** 300*300 Google Earth Engine was used to obtain high-resolution satellite images of Uganda, with a focus on the capital city due to memory size limitations. This limitation restricted the scope of the dataset, meaning the model was trained primarily on images from the capital, which could affect the generalizability of the results to the entire country.
- **Hospital Data:** The dataset for hospital locations in Uganda was sourced from reliable platforms like the World Bank, providing ground-truth data for validating the model's accuracy. **Non-hospital data**, such as restaurants, schools, and other amenities, were downloaded using OpenStreetMap. Due to memory constraints, the dataset was limited to the capital city, which may introduce bias into the model's prediction

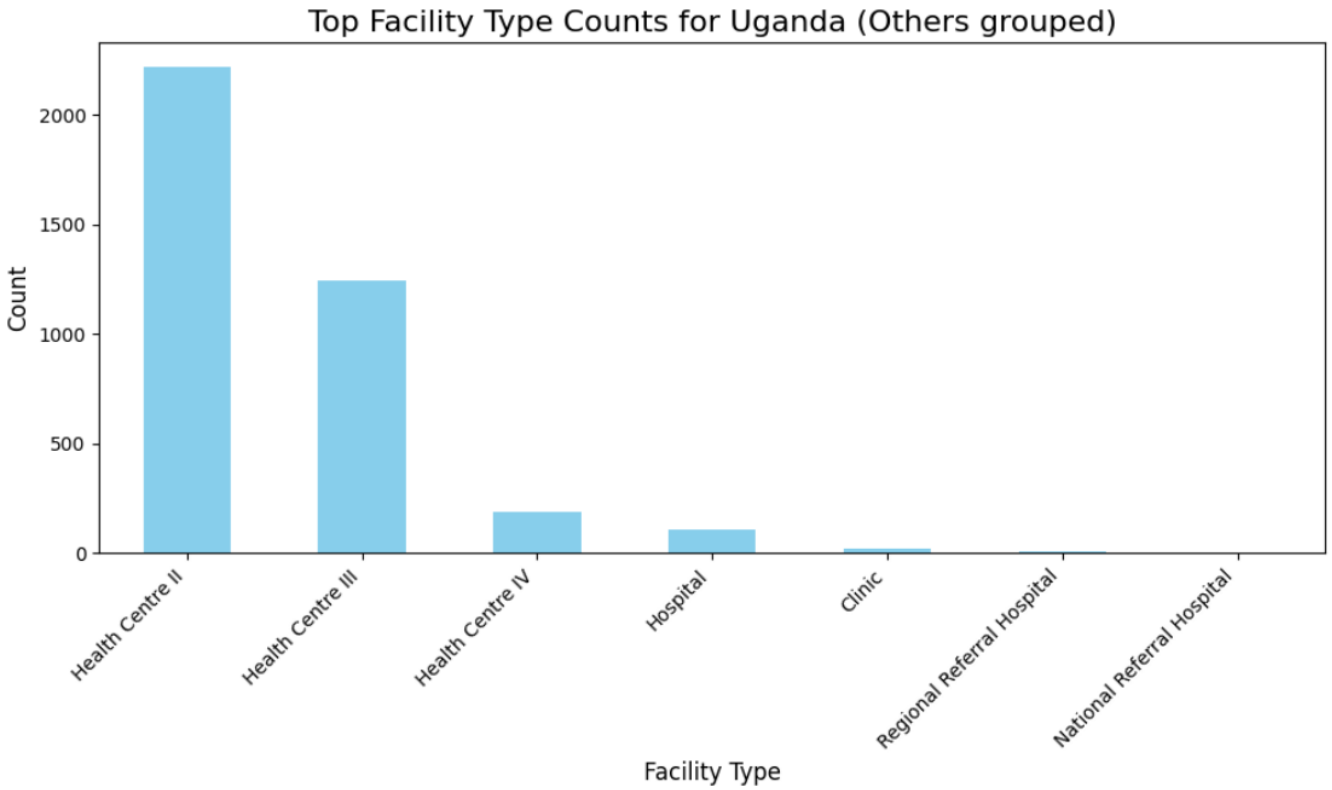
Challenges Encountered in Data Collection

- **Accessing MABPOX Data:**
- **Obstacle:** One of the significant challenges was gaining access to data from MABPOX (Mapping and Bridging the Poverty and Opportunities Exchange), particularly because I wasn't granted access to the website.
- **Solution:** To overcome this, I turned to alternative sources like Google Earth and Eco Browser. However, I found Google Earth Engine (GEE) to be more flexible and reliable for acquiring satellite imagery, providing better tools and features for the project.
- **Google Earth Engine Data Format and Size Issues:**
  - **Obstacle:** Google Earth Engine (GEE) provides satellite imagery in TIFF (Tagged Image File Format), which, by default, results in very large file sizes.
  - **Issue:** Initially, I was unfamiliar with managing these large files. The large sizes posed storage and processing challenges, especially when handling multiple high-resolution images. A key complication arose when attempting to download data in the GEE interface, which defaulted to TIFF format, leading to difficulties in efficiently storing and manipulating the images.
  - **Impact on Geo-Referencing:** One major issue with downloading these TIFF files was the difficulty in setting the images as JPEGs without losing geo-referencing data. Geo-referencing is essential for accurate location mapping, as satellite images must have precise geographic coordinates to align with the hospital locations. When trying to set these files as JPEGs, I experienced a loss of geo-referencing information. This caused challenges in maintaining the spatial accuracy necessary for the model to correctly identify and map hospitals within Uganda's geographical boundaries.

Data Preprocessing:

- **Cleaning Data Header Format:**
  - Initially, the dataset was cleaned by addressing issues related to the header format. The raw data often contained inconsistent or incorrect headers, which were adjusted to ensure compatibility with analysis tools.
- **Converting Data to DataFrame:**
  - After cleaning the header, the data was then converted into a structured format using a DataFrame. This allowed for easier manipulation and analysis, as DataFrames are better suited for data processing and exploration.
- **Exploratory Data Analysis (EDA):**
  - Once the data was in a proper format, I conducted exploratory data analysis (EDA) to better understand the dataset. This involved inspecting the hospital and non-hospital data, identifying patterns, and exploring correlations between various features.
  - During this process, I discovered key characteristics in the data that helped refine the subsequent steps of model training, such as the spatial distribution of hospitals and non-hospitals across Uganda.

Uganda has a diverse healthcare system, including hospitals, clinics, health centers, and pharmacies. However, not all facilities are equipped to handle major medical challenges.



Deep Learning Model:

- **Model Training and Preprocessing:** The model was trained using Convolutional Neural Networks (CNNs), which are ideal for image classification tasks due to their ability to learn spatial hierarchies in visual data.
- However, the dataset was imbalanced, with only 121 hospital images compared to 1,540 non-hospital images. This imbalance could have led to biased predictions, as the model might have favored non-hospital images. To address this, several preprocessing steps were implemented:
- **Hospital Image Augmentation:** For the hospital images, data augmentation techniques were applied, such as rotation and rescaling, to artificially increase the dataset and improve the model's ability to generalize.
- **Non-Hospital Undersampling:** To balance the dataset, non-hospital images were undersampled, reducing their number to match that of the hospital images.
- **Image Normalization:** All images were normalized to ensure that the pixel values were within a consistent range, improving model convergence and performance.
- These preprocessing steps were crucial for ensuring that the model could effectively learn to distinguish between hospitals and non-hospitals without being biased by the imbalance in the dataset.

Code Structure and Workflow:

- **Data Download and Storage:**
  - The data for **hospital** and **non-hospital** images is downloaded and stored directly in a folder named `Mapping_Test` on the local file system.
  - Inside this `Mapping_Test` folder, two additional subfolders are created:
    - **Augmented**: This folder holds the hospital images after they have been augmented (through transformations like rotation, rescaling, etc.).
    - **Undersampled Non-Hospital**: This folder contains the undersampled non-hospital images, ensuring that the number of non-hospital images is equal to that of hospital images.
- **Data Augmentation and Undersampling:**
  - **Hospital Images**: Augmentation is applied to hospital images to create more variation. Operations like rotation, scaling, and other image transformations are performed and saved into the "Augmented" folder.
  - **Non-Hospital Images**: These are undersampled (i.e., a portion of the non-hospital images is randomly selected) to match the number of hospital images, and the undersampled images are saved in the "Undersampled Non-Hospital" folder.
- **Temporary Storage and Final Output:**
  - All images are stored in their respective folders under `Mapping_Test` during the session.
  - While the processing happens in-memory, the resulting datasets are saved to disk in these folders, which can later be used for training, validation, and testing of the model.
  - Given the time constraints, more image processing techniques could be applied, but due to limitations, the focus was on basic augmentation and balancing techniques.

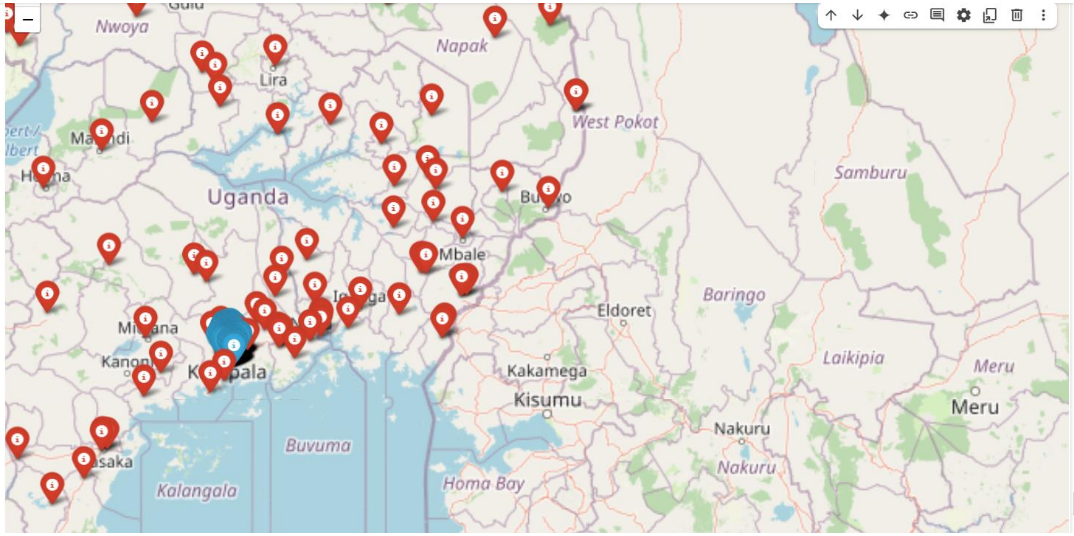
Hospital

Non Hospital

Data:

Mapping Image Locations to Uganda

- Before moving on to model training, I visualized the geographic locations of the hospital and non-hospital images on a map of Uganda. This step helped confirm that the data was accurately aligned with Uganda's geography.
- **Coordinate Extraction:** I extracted the coordinates (latitude and longitude) for both hospital and non-hospital images.
- **Data Organization:** The data was organized into two categories: hospitals and non-hospitals.
- **Mapping:** Using GIS data, I plotted the coordinates onto a map of Uganda. Hospitals were marked in **red**, and non-hospitals in **blue**.
- **Outcome:** This visualization confirmed the correct geographic distribution of the images and highlighted any potential data issues.



Data Splitting for Model Training

split into Train (80%), Validation (10%), and Test (10%) sets.

- To ensure the model learns effectively and generalizes well to unseen data, I divided the dataset into **three subsets**:
- **Training Set:**
  - The largest portion of the data was used for training the model.
  - This set included both hospitals and non-hospitals, ensuring the model had enough data to learn from.
- **Validation Set:**
  - The validation set helped in tuning hyperparameters during the training process.
  - It was used to evaluate the model's performance at each epoch and make adjustments as necessary.
- **Test Set:**
  - The test set was kept separate to assess the model's final performance after training and validation.
  - It provides an unbiased estimate of how the model will perform on unseen data.
  - I used a **stratified split** to maintain the balance between hospitals and non-hospitals across all sets. This ensures that each subset has a similar distribution of the two classes, which helps prevent the model from being biased toward the majority class.

	Dataset Split	Category	Number of Images
0	train	hospital_augmented	1631
1	train	undersample_non_hospital	1461
2	val	hospital_augmented	485
3	val	undersample_non_hospital	354
4	test	hospital_augmented	494
5	test	undersample_non_hospital	335

Convolutional Neural Network (CNN) Model Overview

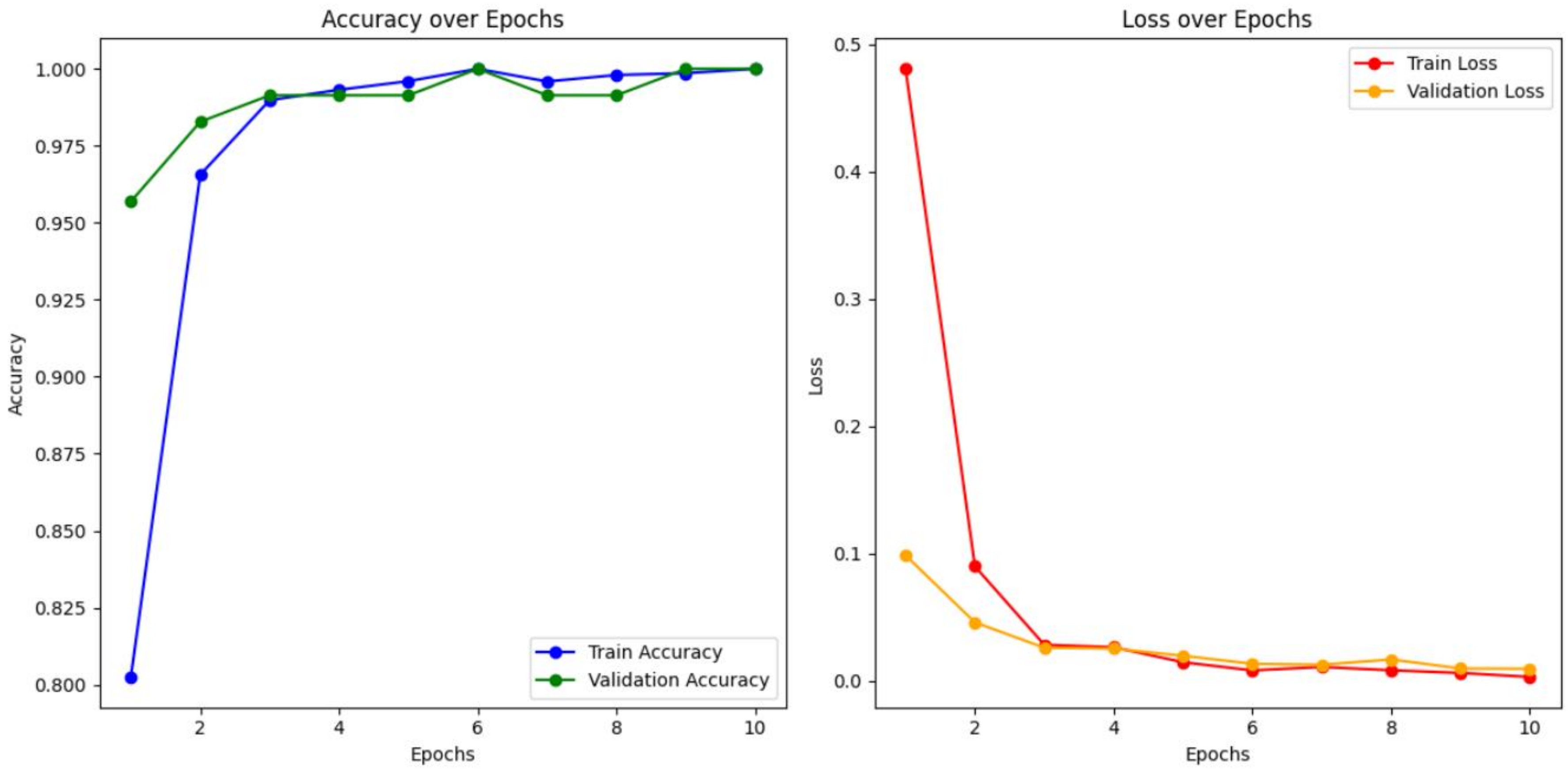
- **Model Configuration:**
- **Input Image Size (244x244)**:
- **Why 244x244?**:
  - **Standard Size**: The image size of 244x244 is a reasonable choice for image classification tasks. It balances performance and computational efficiency.
  - **Resizing**: Images are resized to this size to ensure consistency in the input dimensions. Larger images would require more computation and memory, while smaller ones might lose crucial details.
  - **Aspect Ratio**: This size retains a good aspect ratio while reducing the image dimensions to fit the neural network input.
  - **Performance Optimization**: 244x244 is a typical input size used in many pre-trained models (like VGG16), making it compatible for transfer learning and optimizing model performance.
  - The choice of 244 is close to standard sizes like 224x224 (commonly used in popular models such as VGG and ResNet). The slight increase to 244x244 might be to retain more spatial information for the task at hand.

Training:

- **Optimizer:** Adam, a gradient-based optimizer for faster convergence.
- **Loss Function:** Categorical Cross-Entropy, for multi-class classification.
- **Metrics:** Accuracy, to evaluate model performance.
- **Key Results:**
- **Training Accuracy:** 100%
- **Validation Accuracy:** 100%
- **Loss Decrease:** From 0.48 to 0.0033, indicating successful training.
- **Model Strengths:**
- **Fast Convergence:** Achieved high accuracy in 6 epochs.
- **Generalization:** High validation accuracy suggests the model is not overfitting.

Set Performance:

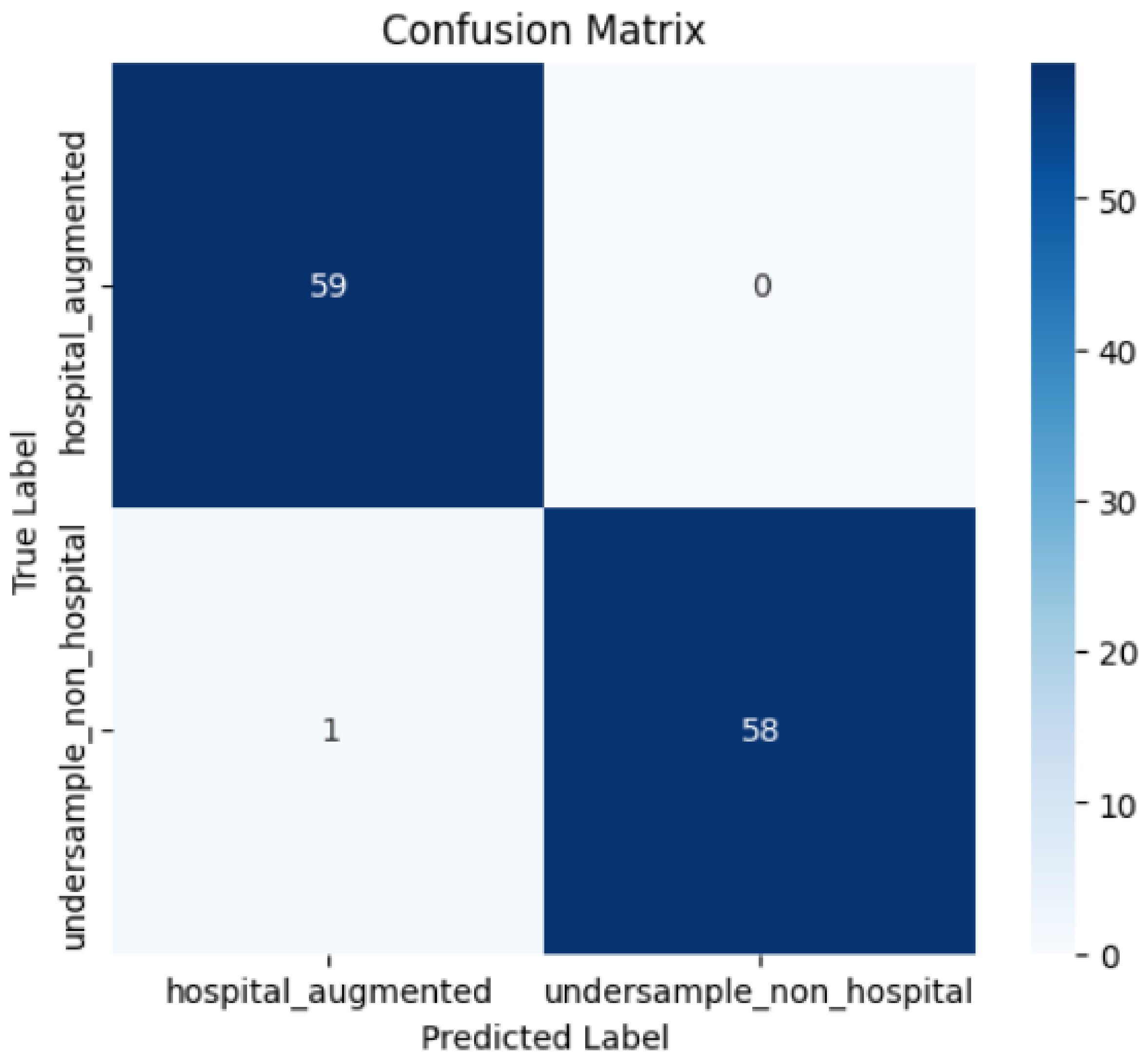
- **Training Accuracy**: The model shows rapid improvement in accuracy, reaching **100% by Epoch 6**.
- **Training Loss**: The loss decreases significantly, indicating the model is learning effectively and reducing prediction errors.
- **Validation Accuracy**: Validation accuracy improves consistently and matches training accuracy by **Epoch 6**, which suggests that the model generalizes well to unseen data.
- **Validation Loss**: Validation loss also decreases, confirming that the model performs better as it learns and does not overfit the training data.



Plot the Metrics

Confusion Matrix Metrics for Each Class:

- **Hospital Augmented (Class 1)**
- **Precision**: 0.98 – Out of all the predicted hospitals, 98% were true positives.
- **Recall**: 1.00 – The model correctly identified 100% of the actual hospitals.
- **F1-Score**: 0.99 – The harmonic mean of precision and recall, indicating a balanced performance.
- **Undersample Non-Hospital (Class 2)**
- **Precision**: 1.00 – Out of all the predicted non-hospitals, 100% were true negatives.
- **Recall**: 0.98 – The model correctly identified 98% of the actual non-hospitals.
- **F1-Score**: 0.99 – A balanced performance between precision and recall.


Confusion Matrix
True Label
hospital_augmented
undersample_non_hospital
59
0
1
58
Predicted Label
hospital_augmented
undersample_non_hospital
50
40
30
20
10
0

Conclusion

- The project successfully demonstrated the end-to-end process of predicting hospitals from satellite imagery using deep learning. By carefully selecting the dataset, cleaning the data, and building a well-structured CNN, we achieved excellent classification performance. The results show the power of deep learning in geospatial applications, particularly in areas like healthcare mapping in Africa.
- Key takeaways:
- The importance of careful data preparation and cleaning to ensure the quality of training data.
- The selection of CNNs as an effective deep learning architecture for image classification.
- The ability to achieve high classification accuracy without the need for paid APIs or GPUs, making this solution accessible and cost-effective.

Closing Remarks

- Thank you for this opportunity. I truly enjoyed diving into remote sensing data and exploring various types of research in sustainable development. This project has not only deepened my understanding of the subject but also enhanced my technical and analytical skills.